

Co-design of Simulation Scenarios for Integration of Smart Glasses in Healthcare Training

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Abstract

Wearable smart glasses have been used to enhance learning experiences through mechanisms such as improving concentration, supporting interview practice, and providing feedback on non-verbal communication. However, the potential benefits of remotely supervised training experiences using smart glasses in healthcare are not yet established. Our in-progress exploratory qualitative study addresses ethical, pedagogical, and practical considerations surrounding the use of smart glasses in the physiotherapy training context. We present a novel use of Vuzix Blade 2.0 smart glasses in healthcare via two simulation-based learning scenarios applied to physiotherapy training. The scenarios were co-designed with stakeholders including physiotherapy students, educators, supervisors, and industry partners. Our study is the first that reveals stakeholder perspectives on the use of smart glasses in physiotherapy training, highlighting key implementation considerations and simulation design requirements. These findings support subsequent pilot testing and the potential for scalable, interdisciplinary application across health education settings more broadly.

Ccs Concepts

Applied computing → Education → Interactive learning environments

Keywords

Smart glasses; Simulation-based learning; Physiotherapy training; Work-integrated learning; Human-computer interaction; Remote supervision; Wearable technology; Health education; Co-design

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1 Introduction

Work integrated learning (WIL) placements enable students to apply theoretical knowledge in real-world healthcare environments, traditionally under the in-person guidance of clinical educators[14]. However, increasing demand for placements driven by rising enrolment and workforce pressures has strained existing capacity, particularly in remote or resource-limited contexts [1, 9, 27]. These challenges call for scalable, pedagogically sound and learner-centred innovations in training delivery. Until now, the use of wearable smart glasses to extend healthcare WIL training opportunities had not been investigated. In other areas of health training, smart glasses have been used to deliver first-person perspectives, enhance learner concentration, support communication training, and facilitate remote feedback [10, 12, 24, 34]. Smart glass technology such as Google Glass and Microsoft HoloLens have shown some feasibility and acceptability in medical education [2], yet their use within healthcare WIL settings remains largely unexplored. Compared to immersive virtual reality systems, smart glasses offer greater usability in clinical contexts due to their familiar appearance, comfort, and lower intrusiveness [2, 3]. Alongside this, simulation-based learning (SBL) is an established strategy in pre-registration healthcare education, offering structured environments for students to practise clinical reasoning, refine behaviours, and engage with realistic scenarios in a safe and controlled setting. SBL provides a scalable model for practice-based learning that is adaptable to various patient presentations and healthcare contexts[21]. Integrating smart glasses into simulation contexts may offer new utility in learning observation and feedback, particularly in preparing students for remote or hybrid placements. Despite the potential benefits, the ethical implications of using smart glasses in healthcare WIL contexts remain poorly understood. Key concerns around privacy, consent, data governance, and equitable access require careful consideration before widespread implementation in the health training setting [2, 4, 15, 30].

To address these gaps, we are conducting an exploratory qualitative study led by a transdisciplinary research team comprising physiotherapy and design specialists. The study involves two key project phases: a co-design workshop and a pilot program, and is structured around three core aims:

- Aim 1: Explore learner, educator, and industry stakeholder perspectives on the ethical use of smart glasses in healthcare WIL settings.
- Aim 2: Co-create two physiotherapy WIL training simulations using Vuzix Blade 2.0 smart glasses.
- Aim 3: Evaluate physiotherapy WIL learner experiences of training simulations that incorporate remote audiovisual supervision and feedback recordings via smart glasses.

This paper reports findings from the completed co-design workshop phase (addressing Aims 1 and 2) and offers a number of contributions in the application of smart glass technology to the healthcare field. We present expert stakeholder perspectives into the feasibility and ethical implications of using smart glasses within healthcare WIL settings. Our findings outline practical design recommendations for incorporating smart glasses into simulation-based learning, including considerations for pre-briefing, supervision, and debriefing protocols. In addition, we present our simulation protocol developed for subsequent pilot testing (towards addressing Aim 3) in the next phase of future implementation and evaluation in our study. To our knowledge, this is the first study to systematically explore the co-design and application of smart glasses technology for physiotherapy placement education.



Figure 1: Participants in the co-design workshop, April 2025.

2 Related Work

2.1 The utility of simulation-based learning in healthcare training

SBL is an established educational approach in physiotherapy and broader healthcare training, particularly for pre-registration students. This method supports the development of clinical knowledge, skills, professional behaviours, and decision-making within a controlled and reflective learning environment [21, 22].

SBL may be designed for diverse experiences ranging across the use of manikins capable of displaying physiological signs, practising discrete clinical workflows, or engaging in fully immersive, long-term simulated placement programs designed to prepare students comprehensively for clinical practice [33]. Physiotherapy students generally perceive SBL positively, noting improvements in skill acquisition and behavioural outcomes [21, 25]. For example, peer-assisted simulation is viewed as beneficial in preparing students for clinical placement despite limitations in realism associated with lower-cost or lower-fidelity scenarios [8]. This highlights SBL's potential for scaling and incorporation into curricula without extensive resource demands. Additionally, SBL provides an intermediate space to test and refine clinical workflows, address ethical and practical concerns, and safely manage potential risks without impacting real patients [26]. This suggests SBL is a useful modality for providing a controlled environment to design, pilot, and improve smart glass integration before wider deployment in WIL placement settings.

2.2 Technical application of smart glasses in physiotherapy training settings

The integration of smart glasses in physiotherapy education is nascent. To our knowledge, only two previous studies have investigated this area. A 2016 study utilizing Google Glass for vestibular training enabled instructors to livestream first-person demonstrations. Although physiotherapy students valued this live perspective, dissatisfaction emerged due to poor video quality, negatively affecting both acceptance and learning outcomes. Nevertheless, students using smart glasses showed improved skill performance compared to controls, emphasizing promising educational potential despite technical limitations [17]. A recent study in 2025 evaluated the efficacy of smart glasses during instruction for the application of neuromuscular electrical stimulation (NMES). Physiotherapy students reported occasional delays in video transmission caused by Wi-Fi connectivity issues, highlighting the necessity for supplementary high-resolution still images. Despite these challenges, visual field sharing via smart glasses resulted in skill mastery comparable to traditional face-to-face demonstrations [29].

Vuzix smart glasses are being increasingly adopted in healthcare training and commercial settings, reflecting their growing role in education and simulation [16, 19, 29]. The Vuzix Blade 2 model was selected for this study due to features aligning with the study requirements for livestreaming and remote supervision: hands-free operation, real-time video capture, Wi-Fi and Bluetooth connectivity and a full-colour, single-eye heads-up display. Additionally, built-in noise-cancelling microphones and stereo speakers support two-way audio communication between supervisors and learners. Lightweight design and adjustable nose pads facilitate extended wear on placements, although regular spectacles cannot be comfortably accommodated under the smart glasses, and require purchase of prescription lens inserts.

3 Methods

3.1 Best practices in simulation design

Towards prototyping of our SBL scenarios, we used a human-centered simulation design approach informed by the INACSL Simulation Standards [31] and the Chartered Society of Physiotherapy Simulation Toolkit [13]. This approach integrates established best practices from adult education, instructional design, clinical standards of care, and simulation pedagogy, prioritizing learner needs and contextual relevance[18]. The recommended process begins with a systematic needs assessment to identify specific gaps in learners' knowledge, skills, and attitudes, ensuring alignment of simulation scenarios with clearly defined learning objectives and addressing discrepancies between current and desired skills performance [20, 22, 31, 32]. Realistic scenarios with suitable fidelity should then be designed, supported by structured pre-briefing and learner-centered facilitation plans that recognize participants' prior knowledge and experience. Finally, reflective debriefing sessions, systematic evaluation strategies, and pilot testing are recommended to ensure effectiveness before implementation into health training curricula [13, 31].

3.2 Simulation scenario prototypes

As recommended above, a needs assessment was first undertaken to inform the development of our SBL scenarios prior to the co-design workshop. This process involved consultation with simulation researchers and experts in healthcare education and pedagogy. Key stakeholders, including clinical educators, senior physiotherapists, and WIL coordinators, were also engaged to identify priority training needs, reflecting the practical competencies required in clinical settings. Based on these combined inputs, two SBL scenarios were selected for further development, co-creation, and refinement during the co-design workshop. These were a clinical handover scenario, and a bed mobility assessment scenario.

3.3 Co-design workshop

The co-design workshop served the first two aims of this study: exploring stakeholder perspectives on the ethical use of smart glasses in healthcare WIL settings, and co-creating two physiotherapy training simulations using Vuzix Blade 2.0 smart glasses. Twenty-two adult participants were purposively sampled for the co-design workshop (Swinburne University HREC ethics approval 20248362-20257). These participants comprised clinical physiotherapist supervisors, educators in design and physiotherapy, and specialists in design and industrial engineering drawn from professional networks, as well as Master of Physiotherapy students from Swinburne University of Technology. Inclusion criteria required participants to have a professional interest or stake in WIL, physiotherapy placement training, and/or the application of wearable technologies within healthcare training environments. The 2.5-hour hybrid workshop was conducted with in-person and online participants. Data collected included audiovisual recordings, written feedback, and

participant-generated artefacts. We structured the workshop following participatory co-creation and human-centred design thinking methodologies, engaging end users early in the design process [11]. The workshop activities were constructed to elicit collective thinking, collaboration, and shared decision-making among stakeholders [28]. Activities began with structured brainstorming using paper and online Miro whiteboards, where stakeholders identified opportunities and challenges of using smart glasses in WIL. Participants also ideated novel applications beyond current practice. The workshop concluded with SBL refinement, focusing on integration of smart glasses into the two simulation scenario prototypes. Participants addressed pre-briefing content, training needs, ethical considerations (e.g., privacy, consent), indicators of learning, and strategies for managing technical issues. Participants reflected on the requirements for remote supervision and debriefing to capture student experiences and feedback. Upon completion of the workshop, a follow-up survey was conducted to gather additional feedback and insights. 2 coders used a thematic analysis approach for data analysis [5], beginning with open sentence-level coding and independent theme identification. This was focused on applying preliminary findings to the simulation design and implementation for the pilot phase.

4 Co-Design Workshop Findings

Contributing to our findings for Aim 1, stakeholders shared useful insights on the integration of smart glasses technology in SBL and WIL contexts, highlighting a range of novel affordances and challenges across the following themes:

4.1 Enhancing learner experiences in healthcare training

Participants recognised the potential of smart glasses to support healthcare educators and students by enabling remote observation and feedback, reducing travel demands, and increasing accessibility. The hands-free format was described as beneficial, allowing students to engage in clinical tasks without impedence. The technology was perceived to “supplement traditional methods of teaching” and foster “more student autonomy.” Real-time feedback was especially valued, with the “ability to get realtime feedback from a supervisor” seen as a significant potential enhancement to learning opportunities.

4.2 Usability of smart glasses in clinical settings

Stakeholders identified potential usability barriers for implementation such as poor “battery life,” interface complexity, and lag in video streaming. Head-mounted camera movement was noted as potentially disorienting for remote viewers. Users wearing prescription glasses raised concerns about comfort and the additional cost of prescription inserts. Broader feasibility challenges included expense, theft risk, and uncertainty about “repair and replace logistics,” particularly for smaller healthcare providers.

4.3 Ethical considerations

Privacy was identified in the workshop as a central concern, especially relating to capturing patients or staff in the background of livestreaming, and around questions over “who keeps data” and “who has access.” Consent processes were critically examined, with some suggesting that “informed consent at timely points [is] better than a blanket rule at the start.” These concerns reflected broader apprehension around data governance and cyber security of smart glasses technology, and the need to address this in the design process.

4.4 Impact on Skill Development

While smart glasses were considered less intrusive than bulkier virtual reality headset alternatives, participants noted they may “reduce quality of social relationships” and heighten “student anxiety.” Being recorded could alter behaviour on placements, with implications for assessment reliability. Nonetheless, supervisors emphasised the value of recorded footage to “visually show errors” and prompt reflection, supporting learning without “micromanaging.” This was an important finding around the level of instruction to offer learners in the SBL design.

5 Application of Findings to SBL Design

The findings from the co-design workshop have directly informed the SBL design to be used in our pilot phase, taking place in 2025 (Aim 3). The integration of smart glass technology within our initial simulation scenario prototype prioritises learner experience, ethical practice, and technical feasibility as follows:

5.1 Learning Objectives

A key workshop finding was the importance of informed consent, transparency, and privacy in clinical settings. Accordingly, the primary learning objective for both SBL scenarios is to apply ethical principles in the use of smart glasses during clinical practice, with specific focus on privacy, consent, and transparency.

This objective is demonstrated through the following learner actions:

- Clearly communicate to patients and healthcare staff when recording or livestreaming is taking place using the smart glasses.
- Obtain informed consent prior to the use of smart glasses, ensuring stakeholders understand the purpose and scope of use.

Additional learning objectives are included, relevant to specific clinical skills supported in each SBL scenario.

5.2 Simulation preparation and pre-briefing

The pilot phase of the study will run across two days at Swinburne University’s high-fidelity simulation lab, which features equipment and design to emulate the environment of an inpatient hospital setting. A trained simulated participant (SP) will portray both staff and patient roles in the scenarios, using detailed scripts

and challenge cues to support consistency and learning outcomes[23]. Physiotherapy students undertaking WIL placements will be invited to participate in the pilot, and will receive pre-briefing outlining learning outcomes, scenarios, case notes, safety and technical procedures. This SBL preparation reflects established best practices as outlined in Section 2.2.

5.3 Simulation Protocol

Participant feedback from the co-design process was instrumental in refining the simulation protocol. Students will wear the Vuzix Blade 2.0 smart glasses during the simulation to livestream their audiovisual perspectives. Supervisors will observe the livestream remotely from an adjacent room and intervene only if technical or safety issues arise. Potential safety considerations include seizure risk, eye strain, motion sickness, and device heat. Verbal protocols will be used to signal the immediate cessation or conclusion of the simulation, and battery packs will supplement the smart glasses where needed. These decisions reflect co-designed input emphasising the importance of minimising distractions, upholding student autonomy, maintaining the therapeutic alliance, and enabling remote support when necessary.

5.4 Debriefing

Immediately following the simulation, students will engage in a 30-minute structured debrief, including a learner experience survey [6]. Students will review their recorded livestreams with supervisors to reflect on performance and receive feedback using the Plus-Delta feedback model [7, 18]. Two recordings from the simulation will be retained with consent for future data analysis and training purposes.

6 Future Work

The next step in this study is the delivery and evaluation of the pilot phase, addressing Aim 3. Findings will inform the potential integration of smart glasses into physiotherapy curricula, support investigation into broader applications across interdisciplinary health training scenarios, and contribute to the development of ethical recommendations for their use in healthcare education settings. Further research is recommended to explore advanced smart glass functionalities in this context, including augmented reality overlays, real-time or on-demand supervisory support, and biometric feedback mechanisms.

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